



Commonwealth of Virginia

VIRGINIA DEPARTMENT OF ENVIRONMENTAL QUALITY

www.deq.virginia.gov

David L. Bulova
Secretary of Natural and Historic Resources

Michael S. Rolband, PE, PWD, PWS Emeritus
Director

March 26, 2026

Mr. Jeffrey J. Davis
Site Director – MAN1
QTS Investment Properties Manassas, LLC
9400 Godwin Drive
Manassas, VA 20110

Location: Prince William County
Registration No.: 74161

Dear Mr. Davis:

Attached is a permit to construct and operate emergency diesel engine generator sets (gen-sets) at QTS Investment Properties Manassas, LLC's data center, in accordance with the provisions of the Virginia State Air Pollution Control Board Regulations for the Control and Abatement of Air Pollution. This combined permit document supersedes your permit document dated October 1, 2024.

In the course of evaluating the application and arriving at a final decision to approve the project, the Department of Environmental Quality (DEQ) deemed the application complete on March 25, 2026.

This permit contains legally enforceable conditions. Failure to comply may result in a Notice of Violation and/or civil charges. Please read all permit conditions carefully.

This permit approval to construct and operate shall not relieve QTS Investment Properties Manassas, LLC of the responsibility to comply with all other local, state, and federal permit regulations.

The proposed emergency diesel engine gen-sets may be subject to the requirements of 40 CFR 60, New Source Performance Standards (NSPS) Subpart IIII – *Standards of Performance for Stationary Compression Ignition Internal Combustion Engines* and 40 CFR 63, National Emission Standards for Hazardous Air Pollutants (MACT) Subpart ZZZZ – *National Emissions Standards for Hazardous Air Pollutants for Stationary Reciprocating Internal Combustion Engines*. In summary, the units may be required to comply with certain federal emission standards and operating limitations. The DEQ advises you to review the referenced MACT and NSPS to ensure compliance with applicable emission and operation limitations. As the owner/operator you are also responsible for any monitoring, notification, reporting, and recordkeeping requirements of the MACT and NSPS. Notifications shall only be sent to EPA, Region 3.

To review any federal rules referenced in the above paragraph or in the attached permit, the US Government Publishing Office maintains the text of these rules at www.ecfr.gov, Title 40, Part 60 and 63.

The Board's Regulations as contained in Title 9 of the Virginia Administrative Code 5-170-200 provide that you may request a formal hearing from this case decision by filing a petition with the Board within 30 days after this case decision notice was mailed or delivered to you. Please consult the relevant regulations for additional requirements for such requests.

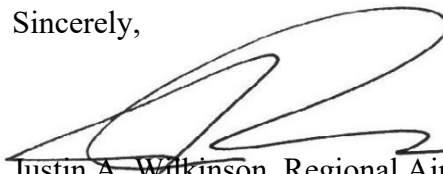
As provided by Rule 2A:2 of the Supreme Court of Virginia, you have 30 days from the date you actually received this permit or the date on which it was mailed to you, whichever occurred first, within which to initiate an appeal of this decision by filing a Notice of Appeal with:

Michael S. Rolband, Director
Department of Environmental Quality
P. O. Box 1105
Richmond, VA 23218

If this permit was delivered to you by mail, three days are added to the 30 day period in which to file an appeal. Please refer to Part Two A of the Rules of the Supreme Court of Virginia for information on the required content of the Notice of Appeal and for additional requirements governing appeals from decisions of administrative agencies.

If you have any questions concerning this permit, please contact Ms. Katie DeVoss at (571) 866-6090 or via email at katie.devoss@deq.virginia.gov.

Sincerely,



Justin A. Wilkinson, Regional Air Permit Manager
Virginia Department of Environmental Quality
(571) 408-1651
justin.wilkinson@deq.virginia.gov
Northern Regional Office
13901 Crown Court, Woodbridge, VA 22193
(703) 583-3800

JAW/KD/74161 mNSR (2026-03-26)

Attachment: Permit



Commonwealth of Virginia

VIRGINIA DEPARTMENT OF ENVIRONMENTAL QUALITY

www.deq.virginia.gov

David L. Bulova
Secretary of Natural and Historic Resources

Michael S. Rolband, PE, PWD, PWS Emeritus
Director

STATIONARY SOURCE PERMIT TO CONSTRUCT AND OPERATE

This permit document supersedes the permit document dated October 1, 2024.

In compliance with the Federal Clean Air Act and the Commonwealth of Virginia Regulations for the Control and Abatement of Air Pollution,

QTS Investment Properties Manassas, LLC
9400 Godwin Drive
Manassas, VA 20110
Registration No.: 74161

is authorized to construct and operate

emergency diesel engine generator sets (gen-sets)

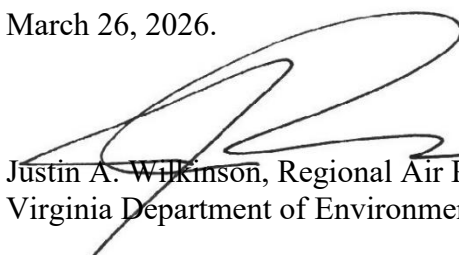
located at

9400, 9420, 9450, and 9540 Godwin Drive;
10680 University Blvd; and
9301 Freedom Center Blvd
Manassas, VA 20110
(Prince William County)

in accordance with the Conditions of this permit.

Approved on

March 26, 2026.


Justin A. Wilkinson, Regional Air Permit Manager
Virginia Department of Environmental Quality

Permit consists of 19 pages (without Attachment).

Permit Conditions 1 to 33.

Attachment A: Source Testing Report Format (1 page)

INTRODUCTION

This permit approval is based on and combines permit terms and conditions in accordance with 9VAC5-80-1255 from the following permit approvals and the respective permit applications:

- mNSR Permit dated March 26, 2026, based on the permit application dated June 25, 2025, including supplemental information dated August 8, 2025, October 27, 2025, December 10, 2025, March 4, 2026, March 20, 2026, and March 25, 2026.
- mNSR Permit dated October 1, 2024, based on the permit application dated July 24, 2024, including supplemental information dated August 19, 2024, and September 23, 2024;
- mNSR Permit dated November 14, 2022, based on the permit application dated August 3, 2022, including supplementation information dated November 10, 2022;
- mNSR Permit dated April 26, 2021, based on the permit application dated December 3, 2020, including supplemental information dated March 25, 2021, and April 2, 2021; and
- mNSR Permit dated December 7, 2018, based on the permit application dated June 4, 2018, including supplemental information dated August 21, 2018, August 28, 2018, September 21, 2018, October 26, 2018, November 5, 2018, and November 29, 2018.

Any changes in the permit application specifications or any existing facilities, which alter the impact of the facility on air quality, may require a permit. Failure to obtain such a permit prior to construction may result in enforcement action. In addition, this facility may be subject to additional applicable requirements not listed in this permit.

Words or terms used in this permit shall have meanings as provided in 9VAC5-80-1110 and 9VAC5-10-10 of the Commonwealth of Virginia's State Air Pollution Control Board Regulations for the Control and Abatement of Air Pollution. The regulatory reference or authority for each condition is listed in parentheses () after each condition. The enabling permit program, or permit programs is provided below each permit condition in the regulatory authority parenthetical as follows: 9VAC5-80-850 for Article 5, 9VAC5-80-1180 for Article 6, 9VAC5-80-1985 for Article 8, and 9VAC5-80-2050 for Article 9. The most recent effective date for a term or condition is listed in brackets []. When identical conditions for one or more emission units are combined, the effective date listed in this permit does not alter the prior effective date(s) for any such conditions as issued in a previous permit action. In accordance with 9VAC5-80-1120F, any condition not marked as state-only enforceable (SOE) is state and federally enforceable.

Annual requirements to fulfill legal obligations to maintain current stationary source emissions data will necessitate a prompt response by the permittee to requests by the Department of Environmental Quality (DEQ) or the Board for information to include, as appropriate: process and production data; changes in control equipment; and operating schedules. Such requests for information from the DEQ will either be in writing or by personal contact.

The availability of information submitted to the DEQ or the Board will be governed by applicable provisions of the Freedom of Information Act, §§ 2.2-3700 through 2.2-3714 of the Code of Virginia, § 10.1-1314 (addressing information provided to the Board) of the Code of Virginia, and 9VAC5-170-60 of the State Air Pollution Control Board's Regulations. Information provided to federal officials is subject to appropriate federal law and regulations governing confidentiality of such information.

Equipment List – Equipment at this facility subject to the permit requirements of 9VAC5-80-1100 *et seq.* consists of:

Equipment to be Constructed:

Ref. No(s).	Equipment Description	Standby Rated Capacity (each unit)	Delegated Federal Requirements	Original Permit Date
Gen138 & Gen159	Two (2) Caterpillar C32 emergency diesel engine gen-sets	1,829 bhp 1,250 ekW	None	March 26, 2026

Equipment Previously Permitted:

Ref. No(s).	Equipment Description	Standby Rated Capacity (each unit)	Delegated Federal Requirements	Original Permit Date
Gen01 – Gen20	Twenty (20) MTU 16V4000G83L emergency diesel engine gen-sets	3,350 bhp 2,250 ekW	None	December 7, 2018
Gen21 – Gen52	Thirty-two (32) MTU 20V4000G64S emergency diesel engine gen-sets	3,672 bhp 2,500 ekW	None	April 26, 2021
Gen53	One (1) MTU 12V4000G24S emergency diesel engine gen-set (Gen-set Model 12V4000 DS1750, Data Center Continuous Power)	2,328 bhp 1,600 ekW (de-rated from 2,561 bhp 1,750 ekW)	None	April 26, 2021*
Gen54	One (1) Caterpillar C32 emergency diesel engine gen-set	1,829 bhp 1,250 ekW	None	October 1, 2024
Gen55 – Gen137 & Gen139 – Gen158	(103) MTU Model 16V4000G84S and/or Caterpillar 3516C emergency diesel engine gen-sets	3,353 bhp 2,250 ekW 3,633 bhp 2,500 ekW	None	November 14, 2022

* See 2026 EA for history of this unit's ekW rating.

Specifications included in the above tables are for informational purposes only and do not form enforceable terms or conditions of the permit.

PROCESS REQUIREMENTS

1. **Emission Controls** – Emissions from the emergency diesel engine gen-sets (Ref. Nos. Gen01 – Gen159) shall be controlled by the following:
 - a. Nitrogen oxides (NO_x) emissions from the emergency diesel engine gen-sets (Ref. Nos. Gen01 – Gen52) shall be controlled by electronic fuel injection and after cooled engines. The permittee shall maintain documentation that demonstrates the control devices have been installed on each emergency diesel engine gen-set.
 - b. Nitrogen oxides (NO_x) emissions from each emergency diesel engine gen-set (Ref. Nos. Gen53 – Gen159) shall be controlled by engine design.
 - c. Carbon monoxide (CO) emissions, particulate matter (PM₁₀ and PM_{2.5}) emissions, volatile organic compounds (VOC) emissions, and visible emissions from the emergency diesel engine gen-sets (Ref. Nos. Gen01 – Gen159) shall be controlled by the use of good operating practices and performing appropriate maintenance in accordance with the manufacturer recommendations. In addition, the permittee may only change those settings that are permitted by the manufacturer and does not degrade the air emissions from the emergency diesel engine gen-sets.

(9VAC5-80-1180 and 9VAC5-50-260) [3/26/2026]
2. **Engine Electrical Power Output** – The engine gen-set (Ref. No. Gen53) shall be equipped with a controller to limit its electrical power output to no more than 1,600 ekW. The controller shall be programmed to initiate a shutdown timer of no more than 60 seconds, if the engine gen-set exceeds its capacity limit. The engine gen-set shall also be equipped with a device to monitor and record its kilowatt output while the unit is operational.

(9VAC5-80-1180 and 9VAC5-50-260) [3/26/2026]
3. **Monitoring Devices** –
 - a. Fuel Flow: Each engine gen-set (Ref. Nos. Gen 01 – Gen159) shall be equipped with a device to continuously measure and record individual fuel consumption (in gallons) for each engine gen-set during operation.
 - b. Engine Operating Hours: Each engine gen-set (Ref. Nos. Gen01 – Gen159) shall be equipped with a non-resettable hour meter which measures the duration of time that each engine is operated.

Each monitoring device (as required in a. and b. above) shall be observed by the permittee with a frequency of not less than once each day the engine gen-set is operated. The permittee shall keep a log of these observations.

Each monitoring device shall be installed, maintained, calibrated (as appropriate), and operated in accordance with approved procedures which shall include, as a minimum, the manufacturer's written requirements or recommendations. The details of the monitoring device calibrations are to be arranged with the Air Compliance Manager of DEQ Northern Regional Office (NRO).

Each monitoring device shall be provided with adequate access for inspection and shall be in operation when the engines are operating.
(9VAC5-80-1180 D and 9VAC5-50-20 C) [3/26/2026]

OPERATING LIMITATIONS

4. **Operation of the Engine-Generation Sets** – The permittee shall operate and maintain each emergency diesel engine gen-set (Ref. Nos. Gen01 – Gen159) and control devices according to the manufacturer's written instructions or procedures developed by the permittee that are approved by the engine manufacturer. In addition, the permittee may only change those settings that are permitted by the manufacturer and do not increase air emissions.
(9VAC5-80-1180) [3/26/2026]
5. **Operating Limitations (Ozone Season)** – No emergency diesel engine gen-set (Ref. Nos. Gen01 – Gen159) shall be operated for scheduled maintenance and readiness testing (Scheduled MCRT), stack testing, or operational training (that involves fuel combustion) between the hours of 7 AM to 5 PM any day during May 1 through September 30. The permittee may petition the DEQ NRO Air Compliance Manager for exceptions to this requirement, with approvals made on a case-by-case basis.
(9VAC5-80-1180) [3/26/2026]
6. **Operating Limitations (Ozone Season) – Integration Operational Period** – During the integration operational period of each diesel engine gen-set (Ref. Nos. Gen21 – Gen159), any operation of the unit (that involves fuel combustion) between the hours of 7 AM to 5 PM any day during the ozone season of May 1 through September 30 shall only occur if the forecast Air Quality index (AQI) for ozone as published on the AirNow website (<https://airnow.gov>) for Northern Virginia for that day is less than or equal to 100. In the event that AirNow-EnviroFlash (www.enviroflash.info) issues an Air Alert for Metropolitan Washington, D.C. for a day which the forecasted AQI for ozone was less than or equal to 100, operation of each unit (which involves fuel combustion) shall be minimized to the maximum extent practical.
(9VAC5-80-1180) [3/26/2026]

7. **Emergency Power Generation** – The emergency diesel engine gen-sets (Ref. Nos. Gen01 – Gen159) shall only be operated in the following modes:
- a. In situations that arise from sudden and reasonably unforeseeable events where the primary energy or power source is disrupted or disconnected due to conditions beyond the control of an owner or operator of a facility including:
 - i. A failure of the electrical grid;
 - ii. On-site disaster or equipment failure; or
 - iii. Public service emergencies such as flood, fire, natural disaster, or severe weather conditions.
 - b. For participation in an Independent System Operator (ISO) declared emergency, where an ISO emergency is:
 - i. An abnormal system condition requiring manual or automatic action to maintain system frequency, to prevent loss of firm load, equipment damage, or tripping of system elements that could adversely affect the reliability of an electric system or the safety of persons or property;
 - ii. Capacity deficiency or capacity excess conditions;
 - iii. A fuel shortage requiring departure from normal operating procedures in order to minimize the use of such scarce fuel;
 - iv. Abnormal natural events or man-made threats that would require conservative operations to posture the system in a more reliable state; or
 - v. An abnormal event external to the ISO service territory that may require ISO action.
 - c. For scheduled maintenance checks and readiness testing (Scheduled MCRT).
 - d. For unscheduled maintenance, testing, and operational training.
 - e. For the integration operational period which is the period of time beginning with the first time the affected units are started on-site and ending when the affected unit is fully integrated with the source's electrical system.

8. **Operating Hours** – The operating hours of the emergency diesel engine gen-sets (Ref. Nos. Gen01 – Gen20) are limited to the following:
- a. No single unit (Ref. Nos. Gen01 – Gen20) shall operate more than 500 hours per year for all purposes (as provided in Condition 7), calculated monthly as the sum of each consecutive 12-month period;
 - b. No single unit (Ref. Nos. Gen01 – Gen20) shall operate more than 100 hours per year for scheduled maintenance checks and readiness testing (Scheduled MCRT) and unscheduled maintenance, testing, and operational training, calculated monthly as the sum of each consecutive 12-month period; and
 - c. The twenty (20) emergency diesel engine gen-sets (Ref. Nos. Gen01 – Gen20) combined shall not operate more than 500 hours per year for scheduled maintenance checks and readiness testing (Scheduled MCRT) calculated monthly as the sum of each consecutive 12-month period. The permittee may petition the DEQ NRO Regional Air Compliance Manager for approval of additional hours to be used for Scheduled MCRT; such requests will be evaluated on a case-by-case basis.

Compliance for the consecutive 12-month period shall be demonstrated monthly by adding the total for the most recently completed calendar month to the individual monthly totals for the preceding 11 months.
(9VAC5-80-1180) [11/14/2022]

9. **Operating Hours** – Each emergency diesel engine gen-set (Ref. Nos. Gen21 – Gen159) shall operate no more than 30 hours per year for scheduled maintenance checks and readiness testing (Scheduled MCRT, as provided in Condition 7.c) and no more than 500 hours per year for all purposes (as provided in Condition 7) combined.

The annual limits for hours of operation shall be calculated monthly as the sum of each consecutive 12-month period. Compliance for the consecutive 12-month period shall be demonstrated monthly by adding the total for the most recently completed calendar month to the individual monthly totals for the preceding 11 months.
(9VAC5-80-1180) [3/26/2026]

10. **Diesel Fuel Throughput Limit** –

- a. The emergency diesel engine gen-sets (Ref. Nos. Gen01 – Gen52 and Gen54 – Gen159), combined, shall consume no more than 675,755 gallons of diesel fuel oil per year, calculated daily as the sum of each consecutive 365-day period for all purposes (as provided in Condition 7).

- b. The emergency diesel engine gen-set (Ref. No. Gen53) shall consume no more than 2,783 gallons of diesel fuel oil per year, calculated daily as the sum of each consecutive 365-day period for all purposes (as provided in Condition 7).

Compliance for the consecutive 365-day period shall be demonstrated daily by adding the total for the most recently completed calendar day to the individual daily totals for the preceding 364 days.

(9VAC5-80-1180) [3/26/2026]

11. **Fuel Specification** – The approved fuel for the emergency diesel engine gen-sets (Ref. Nos. Gen01 – Gen159) is diesel fuel. The diesel fuel shall meet the American Society for Testing and Materials (ASTM) D975 specification for Grade No. 1-D S15 or Grade No. 2-D S15 and have a maximum sulfur content of 15 ppm, per shipment. Exceedance of these specifications may be considered credible evidence of the exceedance of emission limits. A change in the fuel may require a permit to modify and operate.
(9VAC5-80-1180 and 9VAC5-50-260) [3/26/2026]
12. **Fuel Certification** – The permittee shall obtain a certification from the fuel supplier with each shipment of diesel fuel. Each fuel supplier certification shall include the following:
 - a. The name of the fuel supplier;
 - b. The date on which the diesel fuel was received;
 - c. The quantity of diesel fuel delivered in the shipment;
 - d. A statement that the diesel fuel oil complies with the American Society for Testing and Materials specifications (ASTM D975) for S15 diesel fuel oil; and
 - e. The sulfur content of the diesel fuel.

Alternatively, the permittee must obtain approval from the Regional Air Compliance Manager of the DEQ's NRO if other documentation will be used to certify the diesel fuel type.

Fuel sampling and analysis, independent of that used for certification, as may be periodically required or conducted by DEQ may be used to determine compliance with the fuel specifications stipulated in Condition 11. Exceedance of these specifications may be considered credible evidence of the exceedance of emission limits.

(9VAC5-80-1180) [3/26/2026]

EMISSION LIMITS

13. **Emission Limits (Hourly)** – Emissions from the operation of each emergency diesel engine gen-set (Ref. Nos. Gen01 – Gen54, Gen138, and Gen159) shall not exceed the limits specified below:

Pollutant	Ref. Nos. Gen01 – Gen20	Ref. Nos. Gen21 – Gen52	Ref. No. Gen53	Ref. Nos. Gen54, Gen138, & Gen159
Nitrogen Oxides (as NO ₂)	43.0 lb/hr	48.60 lb/hr	29.20 lb/hr	24.09 lb/hr
Carbon Monoxide (CO)	6.6 lb/hr	11.84 lb/hr	5.00 lb/hr	2.10 lb/hr
Volatile Organic Compounds (VOC)	1.32 lb/hr	2.27 lb/hr	1.37 lb/hr	0.40 lb/hr
Particulate Matter (PM)	0.59 lb/hr	0.94 lb/hr	0.39 lb/hr	0.24 lb/hr
Particulate Matter (PM ₁₀)	0.73 lb/hr	1.12 lb/hr	0.46 lb/hr	0.27 lb/hr
Particulate Matter (PM _{2.5})	0.73 lb/hr	1.12 lb/hr	0.46 lb/hr	0.27 lb/hr

Compliance with these emission limits shall be based on the proper operation and maintenance of the emergency diesel engine gen-sets or by testing, if required.
(9VAC5-80-1180 and 9VAC5-50-260) [3/26/2026]

14. **Emission Limits (Hourly)** – Emissions from the operation of each emergency diesel engine gen-set (Ref. Nos. Gen55 – Gen137 and Gen139 – Gen158) shall not exceed the limits specified below:

Pollutant	Caterpillar 3516C	MTU 16V4000G84S
Nitrogen Oxides (as NO ₂)	48.06 lb/hr	44.35 lb/hr
Carbon Monoxide (CO)	6.09 lb/hr	9.37 lb/hr
Volatile Organic Compounds (VOC)	1.21 lb/hr	2.20 lb/hr
Particulate Matter (PM)	0.40 lb/hr	0.78 lb/hr
Particulate Matter (PM ₁₀)	0.59 lb/hr	0.92 lb/hr
Particulate Matter (PM _{2.5})	0.59 lb/hr	0.92 lb/hr

Compliance with these emission limits shall be based on the proper operation and maintenance of the emergency diesel engine gen-sets or by testing, if required.
(9VAC5-80-1180 and 9VAC5-50-260) [3/26/2026]

15. **Emission Limits (Annual)** – Emissions from the combined operation of the emergency diesel engine gen-sets (Ref. Nos. Gen01 – Gen159) shall not exceed the limits specified below:

Pollutant	(Ref. Nos. Gen01 – Gen52 and Gen54 – Gen159)	Facility-Wide (Ref. Nos. Gen01 – Gen159)
Nitrogen Oxides (as NO ₂)	97.73 tpy	98.09 tpy
Carbon Monoxide (CO)	51.04 tpy	51.14 tpy
Volatile Organic Compounds (VOC)	17.88 tpy	17.95 tpy
Particulate Matter (PM)	5.37 tpy	5.37 tpy
Particulate Matter (PM ₁₀)	5.72 tpy	5.73 tpy
Particulate Matter (PM _{2.5})	5.72 tpy	5.73 tpy

These emissions are derived from the estimated overall emission contribution from operating limits. Exceedance of the operating limits may be considered credible evidence of the exceedance of emission limits. Compliance with these emission limits may be determined as stated in Conditions 1, 2, 10, 13, and 14.
(9VAC5-80-1180) [3/26/2026]

16. **Visible Emission Limit** – Visible emissions from each emergency diesel engine gen-set (Ref. Nos. Gen01 – Gen159) shall not exceed 5% opacity except during one six-minute period in any one hour in which visible emissions shall not exceed 10% opacity as determined by EPA Method 9 (reference 40 CFR 60, Appendix A).

During startup and shutdown, visible emissions from each engine gen-set (Ref. Nos. Gen01 – Gen159) shall not exceed 10% opacity except during one six-minute period in any one hour in which visible emissions shall not exceed 20% opacity as determined by EPA Method 9 (reference 40 CFR 60, Appendix A).
(9VAC5-80-1180, 9VAC5-50-260, and 9VAC5-170-160) [3/26/2026]

INITIAL COMPLIANCE DETERMINATION

17. **Stack Test** – Initial performance tests shall be conducted on at least twenty percent (20%) of the units installed for each type of engine model (Caterpillar 3516C and MTU 16V4000G84S) of the 103 emergency diesel engine gen-sets (Ref. Nos. Gen55 – Gen137 and Gen 139 – Gen158) for NO_x (as NO₂) and CO using appropriate EPA reference methods as approved by the Regional Air Compliance Manager of the DEQ's NRO to determine compliance with the emission limits contained in Condition 14.
- a. Emissions testing of each pollutant for each selected emergency diesel engine gen-set shall consist of three (3) one-hour test runs under load. The average of the three (3)

runs shall be reported as the short-term emission rate for that emergency diesel engine gen-set;

- b. Testing shall be performed on the exhaust stack of the emergency diesel engine gen-set to demonstrate compliance with the NO_x and CO emission limits specified in Condition 14. Testing shall be conducted with the emergency diesel engine gen-set operating at ≥ 90 percent of its rated capacity, unless multiple load band testing is approved by DEQ;
- c. Recorded emergency diesel engine gen-set operational information shall include, but not be limited to:
 - i. Generator load/kilowatt output.
 - ii. Fuel consumption and fuel sulfur content of the diesel fuel oil.
- d. Perform testing to demonstrate compliance within 120 days after the integration operational period has commenced, per engine. The integration operational period is defined as: the period of time beginning with the first time the affected unit is started on-site and ending when the affected unit is fully integrated with the source electrical system. If this deadline falls within the ozone season (May 1 through September 30), the facility shall perform testing to demonstrate compliance within 30 days after the end of the ozone season. Tests shall be conducted and reported and data reduced as set forth in 9VAC5-50-30;
- e. The details of the tests are to be arranged with the Regional Air Compliance Manager of DEQ's NRO. The permittee shall submit the test protocol to the Regional Air Compliance Manager of DEQ's NRO, at least 30 days prior to testing to ensure adequate time for DEQ approval. If the test protocol is received by the DEQ with less than 30 days for review and acceptance, DEQ approval may not be issued in a timely manner to allow for testing to take place according to the permittee's schedule;
- f. Should conditions occur which would require rescheduling the testing, the permittee shall notify the Regional Air Compliance Manager of DEQ's NRO, in writing, within seven (7) days of the scheduled test date or as soon as the rescheduling is deemed necessary; and
- g. Two (2) copies (one (1) paper copy and one (1) electronic copy) of the test results shall be submitted to the Regional Air Compliance Manager, DEQ's NRO within 60 days after test completion and shall conform to the test report format enclosed with this permit.

(9VAC5-50-30 and 9VAC5-80-1200) [3/26/2026]

18. **Visible Emissions Evaluation** – Concurrent with the initial performance tests required in Condition 17, Visible Emission Evaluations (VEE) in accordance with 40 CFR Part 60, Appendix A, Method 9, shall also be conducted by the permittee on the emergency diesel engine gen-sets selected for initial performance testing. The details of the tests are to be arranged with the Regional Air Compliance Manager of DEQ's NRO. The permittee shall submit a VEE protocol in conjunction with the initial stack test protocol required by Condition 17, at least 30 days prior to testing.
- a. Should conditions prevent concurrent opacity observations, the Regional Air Compliance Manager of the DEQ's NRO shall be notified in writing, within seven (7) days, and visible emissions testing shall be rescheduled within 30 days. Rescheduled testing shall be conducted under the same operating conditions as the initial performance tests.
 - b. Two (2) copies of the test result (one (1) hard copy and one (1) electronic copy) shall be submitted to the Regional Air Compliance Manager of the DEQ's NRO within 60 days after test completion and shall conform to the test report format enclosed with this permit (Attachment A).

(9VAC5-50-30 and 9VAC5-80-1200) [3/26/2026]

19. **Electrical Power Output Control Device Validation** – An electrical power output control device validation shall be conducted on the engine gen-set (Ref. No. Gen53) to validate that its electrical power output control device (required by Condition 2) prevents the engine gen-set from exceeding its permitted capacity. The validation shall be performed, reported, and demonstrate compliance within 60 days from the date of this permit document (3/26/2026), but in no event later than 120 days after said date. If this deadline falls within the ozone season (May 1 through September 30), the facility shall perform testing to demonstrate compliance within 30 days after the end of the ozone season. The details of the validation are to be arranged with the Regional Air Compliance Manager of the DEQ's NRO. The permittee shall submit a protocol at least 30 days prior to testing. One copy of the test results shall be submitted to the DEQ within 60 days after test completion and shall conform to the test report format enclosed with this permit.
- (9VAC5-80-1180) [3/26/2026]

CONTINUING COMPLIANCE DETERMINATION

20. **Facility Construction** – The emergency diesel engine gen-sets (Ref. Nos. Gen01 – Gen159) shall be constructed so as to allow for emission testing upon reasonable notice at any times, using appropriate methods. This includes constructing the facility/equipment such that volumetric flow rates and pollutant emission rates can be determined by applicable test methods and providing a stack or duct that is free from cyclonic flow. Sampling ports shall be provided when requested at the appropriate locations and safe sampling platforms and access shall be provided.
- (9VAC5-50-30 F and 9VAC5-80-1180) [3/26/2026]

21. **Emission Testing, Visible Emissions Evaluation, and Output Control Device**

Validation – Upon request by the DEQ, the permittee shall conduct stack tests, visible emission evaluations, and/or Output Control Device Validations of the emergency diesel engine gen-sets (Ref. Nos. Gen01 – Gen159) to demonstrate compliance with the emission limits contained in this permit. The details of the tests shall be arranged with the Regional Air Compliance Manager of the DEQ’s NRO.
(9VAC5-80-1200 and 9VAC5-50-30 G) [3/26/2026]

RECORDS

22. **On Site Records** – The permittee shall maintain records of emission data and operating parameters as necessary to demonstrate compliance with this permit. The content and format of such records shall be arranged with the Regional Air Compliance Manager of the DEQ’s NRO. These records shall include, but are not limited to:

- a. Documentation from the manufacturer that each emergency diesel engine gen-set (Ref. Nos. Gen01 – Gen159) is certified to meet EPA Tier 2 emission standards.
- b. Engine information including make, model, serial number, model year, maximum engine (bhp), and engine displacement for each emergency diesel engine gen-set (Ref. Nos. Gen01 – Gen159).
- c. A monthly log of the monitoring device observations as required by Conditions 2 and 3.
- d. The manufacturer’s written operating instructions or procedures developed by the owner/operator that are approved by the engine manufacturer for each emergency diesel engine gen-set (Ref. Nos. Gen01 – Gen159).
- e. Records of the reasons for operation for each emergency diesel engine gen-set (Ref. Nos. Gen01 – Gen159) including, but not limited to: the date, cause of operation, cause of the emergency, the ISO-declared emergency notification, and the hours of operation.
- f. Records, as necessary, to demonstrate compliance with the operating limitations of Condition 5, which includes but is not limited to: times, dates, and reasons for operation of each emergency diesel engine gen-set that was operating between May 1 and September 30.
- g. To verify compliance with Condition 6, maintain records of:
 - i. The forecasted AQI, as determined by the AirNow website for Northern Virginia, for ozone for the day(s) that an emergency diesel engine gen-set operated during the integration operational period;

- ii. The measured AQI, as determined by the AirNow website for Northern Virginia, for ozone for the day(s) that the emergency diesel engine gen-set operated during the integration operational period;
 - iii. Documentation recording any Air Alerts issued for that operating day, as determined by AirNow-EnviroFlash; and
 - iv. Details of commissioning activities, to include, but not limited to, clock hours and duration.
-
- h. Monthly and annual hours of operation of each emergency diesel engine gen-set (Ref. Nos. Gen01 – Gen159) with annual hours of operation calculated monthly as the sum of each consecutive 12-month period.
 - i. Monthly and annual hours of operation of each emergency diesel engine gen-set (Ref. Nos. Gen01 – Gen20) and for all 20 units combined for purposes of Scheduled MCRT, with the annual hours of operation calculated monthly as the sum of each consecutive 12-month period.
 - j. Monthly and annual hours of operation of each emergency diesel engine gen-set (Ref. Nos. Gen21 – Gen159) for purposes of Scheduled MCRT, calculated monthly as the sum of each consecutive 12-month period.
 - k. Daily and annual diesel fuel consumption of each emergency diesel engine gen-set (Ref. Nos. Gen01 – Gen52 and Gen54 – Gen159), calculated daily as the sum of each consecutive 365-day period.
 - l. Combined daily and annual fuel consumption of the emergency diesel engine gen-sets (Ref. Nos. Gen01 – Gen52 and Gen54 – Gen159), with annual fuel consumption calculated daily as the sum of each consecutive 365-day period to verify compliance with the throughput limitation specified in Condition 10.a.
 - m. Daily and annual fuel consumption of the emergency diesel engine gen-set (Ref. No. Gen53), with annual fuel consumption calculated daily as the sum of each consecutive 365-day period to verify compliance with the throughput limitation specified in Condition 10.b.
 - n. Daily and annual emissions calculations for NO_x (as NO₂), CO, VOC, PM, PM₁₀, and PM_{2.5} from the emergency diesel engine gen-sets (Ref. Nos. Gen01 – Gen52 and Gen54 – Gen159), combined, with annual emissions calculated daily as the sum of each consecutive 365-day period.

- o. Daily and annual emissions calculations for NO_x (as NO₂), CO, VOC, PM, PM₁₀, and PM_{2.5} from the emergency diesel engine gen-sets (Ref. Nos. Gen01 – Gen159), combined, with annual emissions calculated daily as the sum of each consecutive 365-day period.
- p. All fuel supplier certifications, as required by Condition 12.
- q. Results of all stack tests, visible emissions evaluations (VEE), and electrical power output control validations.
- r. Scheduled maintenance checks and readiness testing (Scheduled MCRT).
- s. Unscheduled maintenance and operator training, in accordance with Condition 28.

Compliance for the consecutive 12-month period in subsections above (as applicable) shall be demonstrated monthly by adding the total for the most recent month to the individual monthly totals for the preceding 11 months.

Compliance for the consecutive 365-day period in the subsections above (as applicable) shall be demonstrated daily by adding the total for the most recently completed day to the individual daily totals for the preceding 364 days.

These records shall be available for inspection by the DEQ and shall be current for the most recent five years, unless otherwise noted.
(9VAC5-80-1180 and 9VAC5-50-50) [3/26/2026]

NOTIFICATIONS

23. **Initial Notifications** – The permittee shall furnish written notification of the items below to the Air Compliance Manager of the DEQ’s NRO at the following address:

Regional Air Compliance Manager
Department of Environmental Quality
13901 Crown Court
Woodbridge, VA 22193

The permittee shall submit one notification for each building or construction phase containing information on each emergency engine gen-set (Ref. Nos. Gen21 – Gen159) as described below:

- a. The actual date on which installation of the emergency diesel engine generators commenced in the building, or phase, within 30 days after such date. The notification must contain the following:

- i. Name and address of the permittee,
 - ii. The address of the affected source,
 - iii. Unit reference number of the initial unit installed, and
 - iv. The date construction commenced.
- b. The date that the integration operational period started for each emergency diesel engine generator (Ref. Nos. Gen21 – Gen159) within 15 days after the last generator at each building or construction phase completes its integration operational period. If a period of construction is paused or halted for ≥ 45 days this notification shall be provided to the DEQ within 15 days after completion of the integration operational period for the most recently installed engine generator set. The notification must contain the following:
- i. Unit reference number;
 - ii. Engine information including make, model, engine family, serial number, model year, maximum engine power, engine displacement, fuel used;
 - iii. Installation date; and
 - iv. Integration operational period start and end dates.

For the purpose of this notification the integration operational period is defined as: the period of time beginning with the first time the affected unit is started on-site and ending when the affected unit is fully integrated with the sources electrical system.
(9VAC5-50-50 and 9VAC5-80-1180) [3/26/2026]

GENERAL CONDITIONS

24. **Permit Invalidity** – This permit to construct the emergency diesel engine gen-sets (Ref. Nos. Gen01 – Gen52) shall become invalid, unless an extension is granted by the DEQ, if:
- a. A program of continuous construction or modification is not commenced within 18 months from the date of this permit; or
 - b. A program of construction is discontinued for a period of 18 months or more, or is not completed within a reasonable time.

(9VAC5-80-1210)

25. **Permit Invalidation** – This permit to construct the emergency diesel engine gen-sets (Ref. Nos. Gen54 – Gen159) shall become invalid, unless an extension is granted by the DEQ, if:

- a. A program of continuous construction is not commenced within 18 months from the ‘Original Permit Date’ specified in the equipment list in the Introduction section of this permit; or if
- b. A program of construction is discontinued for a period of 18 months or more, or is not completed within a reasonable time.

(9VAC5-80-1210)

26. **Permit Suspension/Revocation** – This permit may be suspended or revoked if the permittee:

- a. Knowingly makes material misstatements in the permit application or any amendments to it;
- b. Fails to comply with the conditions of this permit;
- c. Fails to comply with any emission standards applicable to a permitted emissions unit;
- d. Causes emissions from the stationary source which result in violations of, or interfere with the attainment and maintenance of, any ambient air quality standard;
- e. Fails to operate in conformance with any applicable control strategy, including any emission standards or emissions limitations, in the State Implementation Plan in effect at the time an application for this permit is submitted; or
- f. Fails to comply with the applicable provisions of 9VAC5-80-1100 *et seq.*

(9VAC5-80-1210 F and 9VAC5-80-1210 G)

27. **Right of Entry** – The permittee shall allow authorized local, state, and federal representatives, upon the presentation of credentials:

- a. To enter upon the permittee's premises on which the facility is located or in which any records are required to be kept under the terms and conditions of this permit;
- b. To have access to and copy at reasonable times any records required to be kept under the terms and conditions of this permit or the State Air Pollution Control Board Regulations;

- c. To inspect at reasonable times any facility, equipment, or process subject to the terms and conditions of this permit or the State Air Pollution Control Board Regulations; and
- d. To sample or test at reasonable times.

For purposes of this condition, the time for inspection shall be deemed reasonable during regular business hours or whenever the facility is in operation. Nothing contained herein shall make an inspection time unreasonable during an emergency.
(9VAC5-170-130 and 9VAC5-80-1180)

28. **Maintenance/Operating Procedures** – At all times, including periods of startup, shutdown, and malfunction, the permittee shall, to the extent practicable, maintain and operate the affected source, including associated air pollution control equipment, in a manner consistent with good air pollution control practices for minimizing emissions.

The permittee shall take the following measures in order to minimize the duration and frequency of excess emissions:

- a. Develop a maintenance schedule and maintain records of all scheduled and non-scheduled maintenance;
- b. Maintain an inventory of spare parts;
- c. Have available written operating procedures for equipment. These procedures shall be based on the manufacturer's recommendations, at a minimum; and
- d. Train operators in the proper operation of all such equipment and familiarize the operators with the written operating procedures, prior to their first operation of such equipment. The permittee shall maintain records of the training provided including the names of trainees, the date of training, and the nature of the training.

Records of maintenance and training shall be maintained on site for a period of five years and shall be made available to DEQ personnel upon request.
(9VAC5-50-20 E and 9VAC5-80-1180 D)

29. **Record of Malfunctions** – The permittee shall maintain records of the occurrence and duration of any bypass, malfunction, shutdown or failure of the facility or its associated air pollution control equipment that results in excess emissions for more than one hour. The records shall be maintained in a form suitable for inspection and maintained for at least two years (unless a longer period is specified in the applicable emission standard) following the date of occurrence. Records shall include the date, time, duration, description (emission unit, pollutant affected, cause of malfunction), corrective action, preventive measures taken, and name of person generating the record.
(9VAC5-20-180 J and 9VAC5-80-1180 D)

30. **Notification for Facility or Control Equipment Malfunction** – The permittee shall furnish notification to the Regional Air Compliance Manager of the DEQ’s NRO of malfunctions of the affected facility or related air pollution control equipment that may cause excess emissions for more than one hour. Such notification shall be made no later than four daytime business hours after the malfunction is discovered. The permittee shall provide a written statement giving all pertinent facts, including the estimated duration of the breakdown, within 14 days of discovery of the malfunction. When the condition causing the failure or malfunction has been corrected and the equipment is again in operation, the permittee shall notify the Regional Air Compliance Manager of the DEQ’s NRO.
(9VAC5-20-180 C and 9VAC5-80-1180)
31. **Violation of Ambient Air Quality Standard** – Regardless of any other provision of this permit, the permittee shall, upon request of the DEQ, reduce the level of operation or shut down a facility, as necessary to avoid violating any primary ambient air quality standard and shall not return to normal operation until such time as the ambient air quality standard will not be violated.
(9VAC5-20-180 I and 9VAC5-80-1180)
32. **Change of Ownership** – In the case of a transfer of ownership of a stationary source, the new owner shall abide by any current permit issued to the previous owner. The new owner shall notify the Regional Air Compliance Manager of the DEQ’s NRO of the change of ownership within 30 days of the transfer.
(9VAC5-80-1240)
33. **Permit Copy** – The permittee shall keep a copy of this permit on the premises of the facility to which it applies.
(9VAC5-80-1180)

Attachment A

Source Testing Report Format

SOURCE TESTING REPORT FORMAT

Report Cover

1. Plant name and location
2. Units tested at source (indicate Ref. No. used by source in permit or registration)
3. Test Dates.
4. Tester; name, address and report date

Certification

1. Signed by team leader/certified observer (include certification date)
2. Signed by responsible company official
3. *Signed by reviewer

Copy of approved test protocol

Summary

1. Reason for testing
2. Test dates
3. Identification of unit tested & the maximum rated capacity
4. *For each emission unit, a table showing:
 - a. Operating rate
 - b. Test Methods
 - c. Pollutants tested
 - d. Test results for each run and the run average
 - e. Pollutant standard or limit
5. Summarized process and control equipment data for each run and the average, as required by the test protocol
6. A statement that test was conducted in accordance with the test protocol or identification & discussion of deviations, including the likely impact on results
7. Any other important information

Source Operation

1. Description of process and control devices
2. Process and control equipment flow diagram
3. Sampling port location and dimensioned cross section Attached protocol includes: sketch of stack (elevation view) showing sampling port locations, upstream and downstream flow disturbances and their distances from ports; and a sketch of stack (plan view) showing sampling ports, ducts entering the stack and stack diameter or dimensions

Test Results

1. Detailed test results for each run
2. *Sample calculations
3. *Description of collected samples, to include audits when applicable

Appendix

1. *Raw production data
2. *Raw field data
3. *Laboratory reports
4. *Chain of custody records for lab samples
5. *Calibration procedures and results
6. Project participants and titles
7. Observers' names (industry and agency)
8. Related correspondence
9. Standard procedures

* Not applicable to visible emission evaluations